

\$52m in AI. What is it delivering?

Meridian is a \$6.5bn industrial enterprise. AI is already embedded across eight divisions. Before approving another \$30m, the board asks for an enterprise assessment.

16 slides / Meridian: \$6.5bn annual revenue

AI USAGE

What does AI do in the process?

BUSINESS IMPACT

Does the completed work improve?

ACCOUNTABILITY

What should continue, change, or stop?

“We have AI everywhere. Do we have a strategy?”

The board’s assessment mandate

“Divisions report \$110m in value.
We need one defensible view.”

Each division has selected tools, built models, and reported success differently. Leadership cannot reconcile adoption, total cost, verified outcomes, or responsibility for AI actions.

SCOPE

Deployed AI and active pilots across eight divisions; four process families examined here.

SPEND VISIBILITY

\$36m in the central view; assessment reconciles another \$16m.

PENDING REQUEST

\$30m additional investment, separate from existing annual costs.

DECISION

What to scale, consolidate, remediate, restrict, or stop.

Four process families. Four distinct AI assessments.

MANUFACTURING / \$14M COST

Vision models inspect products.

Check defect detection, missed defects, false rejects, scrap, rework, overrides, and drift after a line or product change.

MAINTENANCE / \$12M COST

Models predict equipment failures.

Check alert precision, useful warning time, technician action, and whether “avoided downtime” has a defensible comparison.

ORDER OPERATIONS / \$10M COST

AI reads, drafts, and invokes tools.

Trace order extraction, customer responses, and credit recommendations through correction, approval, and final posting.

ENGINEERING / \$16M COST

Copilots generate code and retrieve designs.

Inspect actual adoption, review effort, defects, IP boundaries, duplicated licenses, and whether released time changes delivery.

Four families total \$52m, including allocated shared costs.

Evaluate distinct models, users, data, actions, and business outcomes. Shared spending is counted once.

The central budget shows \$36m. The ledger shows \$52m.

Annual cost category	Illustrative cost	Evidence to reconcile
AI licenses and embedded AI subscriptions	\$9m	Contracts, seats, entitlements, active use
Inference, training, and dedicated AI compute	\$11m	Cloud tags, runtime logs, model-level usage
AI applications, plant systems, managed services	\$10m	Supplier invoices and divisional procurement
AI delivery, integration, and ongoing operations	\$16m	Allocated staff and partner delivery costs
AI-specific data, evaluation, security, governance	\$6m	Attributed team costs, tooling, and assurance work

\$36m centrally reported + \$16m embedded in divisional and shared budgets = \$52m. Reconcile accounts and allocations before adding totals; exclude unrelated ERP and general cloud costs.

One portfolio register. Different tests for each AI use.

Evidence stream	What the assessment examines	Accountable reviewers	Decision it supports
E01 · Inventory + cost	Systems, models, vendors, owners, environments, and allocated costs	Finance + IT + Procurement	What AI exists, who pays, and where spending overlaps
E02 · Intended operation	Process maps, model purpose, source access, tools, approval rules	Business + system owners	What the AI is supposed to do and who owns its limits
E03 · Observed behavior	Risk-based output samples, production telemetry, controlled tests	Quality + Security + Operations	Accuracy, human rework, drift, and reachable actions
E04 · Value evidence	Scrap/rework costs, paid services, time studies, and benefit claims	Finance + process owners	Verified cost reduction versus capacity and forecasts
E05 · Adoption + coverage	Observed work, exceptions, bypasses, and division/shift coverage	Assessment + division leads	Whether results are representative enough to scale

An untested deployment stays visible as an evidence gap. A vision model, predictive model, copilot, and tool-using agent cannot share one generic “AI accuracy” score.

“Human approval required” is not an enforced control.

E02 / Approved process

“AI drafts a credit. A person approves it.”

The process owner expects AI to recommend, with approval before posting.

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E03 / Controlled test

The AI service identity can post a credit.

A test bypasses the required approval because the tool permission is too broad.

F01 / The AI action boundary exceeds the approved business process.

Disable the posting capability pending correction. Preserve useful drafting while testing the approval boundary outside the model.

Change the assessment when the evidence changes.

01 / INITIAL VIEW

2/5

Emerging

Evidence confidence: Provisional

Approved process requires a human decision. Actual AI tool permissions have not yet been tested.

02 / RECONCILED VIEW

1/5

Ad hoc

Evidence confidence: Strong

E03 reproduces credit posting without required approval. The AI tool boundary does not enforce the process.

Risk & Governance / assessed credit workflow only

Recorded change: 2 → 1 after permission testing. Strong evidence can support a low maturity finding. Missing evidence is not a score of zero.

What makes \$54m acceptable evidence?

E04 / Illustrative benefit reconciliation

Compare the same work, over the same period.

Use one completed 12-month period. Finance adjusts the comparison for volume, product mix, input prices, and non-AI process changes before accepting a reduction.

E04-A + E04-B → \$54m accepted → D01

E04-A / VISION

\$118m normalized scrap/rework comparison – \$88m observed cost = \$30m reduction.

E04-B / ORDERS

\$96m normalized outsourced-processing comparison – \$72m observed cost = \$24m reduction.

PROOF TO INSPECT

Source ledger, matched sites/workflows, adjustment log, quality results, and named Finance approval.

NO DOUBLE COUNTING

These business costs exclude AI costs. Subtract the \$52m AI cost once, at portfolio level.

The portfolio deserves four different decisions.

AI process family	Annual allocated cost	Accepted cash cost reduction	Assessment recommendation
Manufacturing vision	\$14m	\$30m	Scale selectively after line-specific evaluation and drift checks
Predictive maintenance	\$12m	Not yet established	Validate alert-to-work-order outcomes and downtime attribution
Order operations	\$10m	\$24m	Continue proven assistance; fix extraction and restrict credit posting
Engineering copilots	\$16m	Not yet established	Measure delivery outcomes; consolidate overlap before adding seats

Accepted reductions total \$54m in this fictional case. “Not yet established” means insufficient financial evidence, not proof of zero value. Capacity and resilience may matter separately.

The \$110m headline becomes a different board conversation.

Finance-accepted cash cost reductions	\$54m
Modeled staff capacity - not cash	\$36m
Forecasts - not yet evidenced	\$12m
Duplicate claims removed	\$8m

Divisional headline: \$110m. Only \$54m enters the cash comparison; the remaining categories are separate.

Assessed annual AI cost: \$52m

BENEFIT LESS ALLOCATED ANNUAL AI COST

\$2m

at the assessed \$52m annual AI cost

A 10% benefit shortfall turns \$2m into –\$3.4m at the same cost. Break-even requires 96.3% of the assessed \$54m benefit.

The web version varies cost and benefit retention. This is not project ROI; additional investment and transition costs are excluded.

Scale what survives the right test.

VISION / QUALITY GATE

Test at the production line.

Measure escaped defects and false rejects against a matched production baseline. Retest when the product, camera, lighting, or line changes.

GENERATIVE AI / CONTROL GATE

Inspect the completed work.

Evaluate corrected orders, reviewed code, and resolved enquiries. Test data boundaries, unsafe tool calls, approval bypasses, and fallback procedures.

Define owners, thresholds, samples, and stop rules per use. Broad AI adoption, a successful demo, or one high-performing site does not establish enterprise readiness.

MAINTENANCE / VALUE GATE

Connect alerts to work orders.

Trace precision, lead time, intervention, and outcome. Adjust comparisons for asset condition and scheduled maintenance before crediting avoided downtime.

PORTFOLIO / FUNDING GATE

Accept the incremental case.

Finance and process owners sign the evidence, coverage, costs, and benefit mechanism. New automation authority gets a separate release decision.

Decision D01: release \$4m. Hold the remaining \$26m.

MERIDIAN / BOARD INVESTMENT DECISION

Authorize / \$4m cap

Targeted scale and essential remediation.

\$2m for proven vision use at qualified lines; \$2m for control fixes, outcome measurement, and independent retesting. Gate each release.

Accountable sponsor: CEO / executive investment committee

Financial acceptance: CFO / named benefit owners

EVIDENCE-GATED FUNDING

Hold / \$26m

Broad rollout and unproven new authority.

Restrict unsafe credit posting immediately. Revisit remaining investment with per-use evidence; review the separate \$52m annual cost and renewals.

Operational authority: division owners + CIO / Security

Next review: gates completed, before renewal commitments

The assessment becomes an enterprise operating agenda.

Finding / decision	Required action	Receiving owner	Evidence to close
F01 · AI action boundaries	Restrict credit posting; test approvals and other agent identities	Process owners + Security	Independent negative tests and approval audit trail
F02 · Uneven outcome evidence	Validate maintenance attribution, extraction rework, and engineering delivery	Division leads + Finance	Use-specific outcomes, representative scope, accepted comparison
F03 · Fragmented cost + claims	Reconcile \$52m; remove duplicate benefits; validate \$8m cost hypothesis	Finance + Procurement + CIO	Ledger, allocations, executable terms, retained quality
D01 · Selective investment	Release funds by evidence gate; review the \$26m held balance	Executive investment committee	Accepted findings, incremental economics, accountable owners

The result is a governed portfolio of AI in business processes—not one enterprise-average maturity score or an instruction to cut everything.

A usable record, not just a score.

01 / THE DECISION

Executive assessment memo

Recommendation, alternatives, conditions, and accountable decision roles.

02 / THE BASIS

Evidence + maturity pack

Charter, source register, six-dimension view, confidence, and dated reasoning.

03 / THE LIMITS

Findings + contradiction register

Accepted findings, dissent, missing coverage, and unresolved assumptions.

04 / THE NEXT MOVE

Owned handoff register

Actions, prerequisites, due dates, receiving owners, and review triggers.

The engagement begins with your decision.

Start with a material investment, a stalled initiative, or a portfolio question that leadership needs resolved.

Scope determines the engagement.

Timing and effort depend on access, coverage, complexity, and the decision deadline.

ALPHAWORX BRINGS

Assessment design, evidence reconciliation, independent challenge, and a decision-ready recommendation.

YOUR ORGANIZATION BRINGS

A sponsor, accountable owners, source access, practitioner participation, and decision authority.

WE AGREE TOGETHER

Scope, evidence standard, schedule, deliverables, and acceptance criteria.

[START A CONVERSATION](#)

What decision is your AI program waiting on?

Bring the proposal, the unresolved question, and the people accountable for the outcome.

[Discuss an assessment ↗](#)

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About this walkthrough

Meridian Industrial Group is a fictional \$6.5bn industrial enterprise. All AI budgets, process allocations, benefits, findings, and decisions are constructed examples. Research informs the enterprise context; the figures are not client results or industry benchmarks.